Federal Economist Compensation & PhD Placement: Evidence by Gender

Updated Data and New Sources

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Outline

1. Overview and Data
2. Federal Salary Analysis
3. Comparison to Prior Work
4. The 2025 Update
5. PhD Placement Analysis
6. Summary

Overview

This project extends prior work on gender equity among economists in two ways:

1. Updates the federal salary analysis through FY2024 using FedsDataCenter (previously through FY2015 in Foster, Manzella, McEntarfer & Sandler 2020)
2. Introduces a new data source on PhD economist initial placements (44 programs, 1991–2025) to complement the LEHD-based analysis in Foster, McEntarfer & Sandler (2023)

Key questions:

- How has the federal economist gender pay gap evolved since 2015?
- How do initial placement shares compare across sectors and over time?
- How does the gender composition of placements compare to the full SED/LEHD sample?

Data Sources

Federal salary data

- Source: FedsDataCenter.com (OPM data via FOIA)
- Years: FY2015–FY2024
- ≈4,300–4,700 economists per year
- Occupation code: "Economist"
- Fields: name, agency, grade, pay plan, salary, bonus
- Gender assigned via `gender_guesser` (offline, ≈84% coverage)

PhD placement data

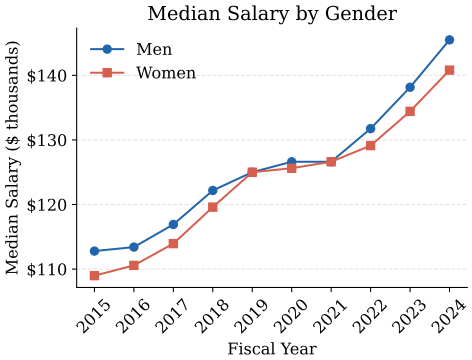
- Source: econphdplacements.com
- 8,824 placements from 44 programs
- Includes US and international programs (e.g., Stockholm, UBC, LSE)
- Years: 1991–2025
- Fields: name, year, program, field, placement, category
- Gender assigned via `gender_guesser` (≈51% coverage)

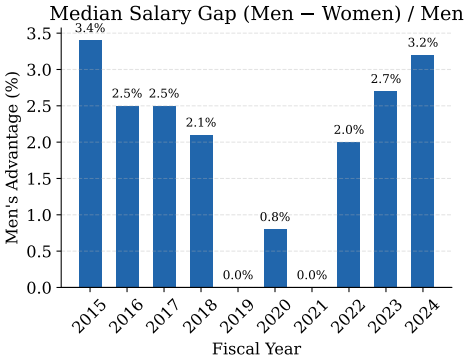
A Note on Gender Assignment

Both datasets use **name-based gender assignment** via `gender_guesser` (an offline library covering ≈40,000 names):

- Returns: *male*, *mostly_male*, *female*, *mostly_female*, *androgynous*, *unknown*
- We assign male/female from the first two; leave the rest unassigned
- Salary data: **84%** coverage (OPM names are mostly Anglo-American)
- Placement data: **51%** coverage (many international names not in library)
- Unassigned names skew international and likely male-heavy ⇒ female shares in placement data are *understated*
- Prior papers (Foster et al. 2020, 2023) use **self-reported** SED gender (100% coverage, no bias)

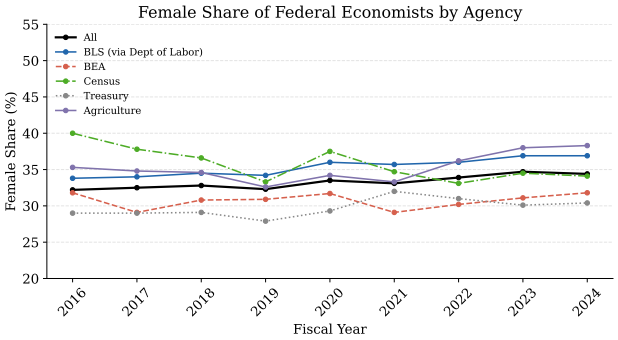
Median Salary by Gender, FY2015–2024





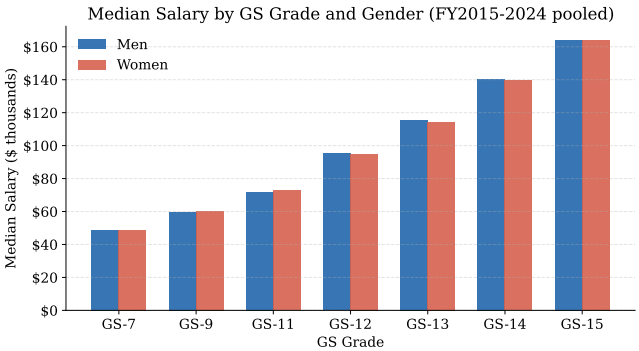
- Gap peaked at 3.4% in 2015, nearly closed in 2019–2021, then **re-widened to 3.2%** by 2024
- 2024 medians: men $145,489 women $140,807

Female Share of Federal Economists by Agency



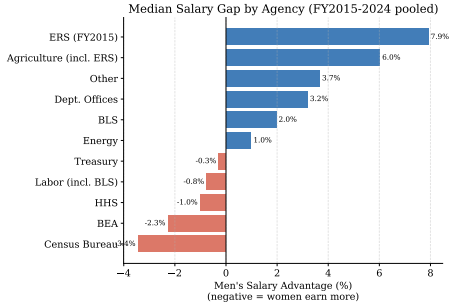
- Overall share rising: 32% (2016) $\rightarrow$ 34% (2024)
- HHS consistently highest ($\approx$40–45%); Treasury lowest ($\approx$29–32%)

Is the Gap Within or Between Grade?



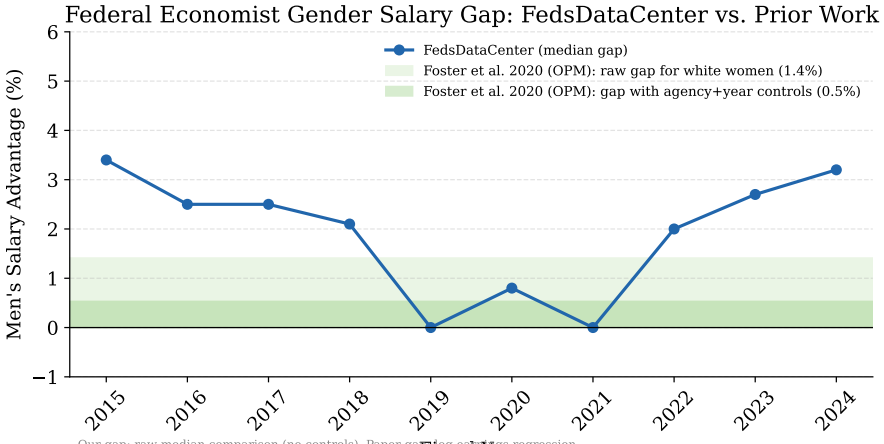
- Within-grade gaps are **near zero or reversed** across all GS levels
- The overall salary gap is driven by **grade distribution**: women are concentrated at lower GS grades

Salary Gap by Agency



- BEA, Census Bureau, and BLS show female salary advantage (women earn more at median)
- Largest male advantage: Agriculture/ERS and "Other" agencies
- Pattern consistent with within-grade gaps being near zero: agency differences reflect grade composition

Federal Salary Gap: FedsDataCenter vs. Foster et al. (2020)



Federal Salary Gap: Comparison Summary

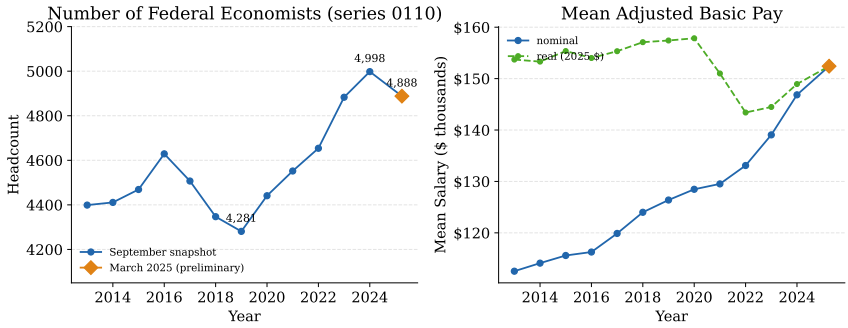
Source	Sample	Specification	Gap
FedsDataCenter (FY2015)	All GS economists	Raw median	3.4%
FedsDataCenter (FY2024)	All GS economists	Raw median	3.2%
FedsDataCenter (FY2019–21)	All GS economists	Raw median	≈0%
Foster et al. (2020)	PhD economists, OPM	Raw log gap (white women)	1.4%
Foster et al. (2020)	PhD economists, OPM	Agency + year FE	0.5%
Foster et al. (2023)	PhD economists, LEHD	Raw log gap, government	8.6%
Foster et al. (2023)	PhD economists, LEHD	Full controls, government	3.8%

- Our raw 3% median gap vs. OPM-based 1.4% raw log gap: broadly consistent
- LEHD-based 8.6% is larger because it is **PhD-only** and captures broader compensation (not just base salary)

The 2025 Update: Data

- Question: has the **number** and **pay** of federal economists changed into 2025?
- FedsDataCenter stops at FY2024. 2025 uses **OPM FedScope**: preliminary **March 2025** snapshot with **September 2024** bundled for comparison (occupational series 0110).
- Three caveats up front:
  - **No gender / race** — FedScope's raw files never had them (the interactive site did, until removed in March 2025). This update is headcount and pay only.
  - **Preliminary**, and still counts administrative-leave / deferred-resignation staff as employed $\Rightarrow$ headcount loss understated.
  - Different universe from FedsDataCenter (EHRI month-end status); small agency $\times$ grade cells suppressed ($\sim$76% coverage).

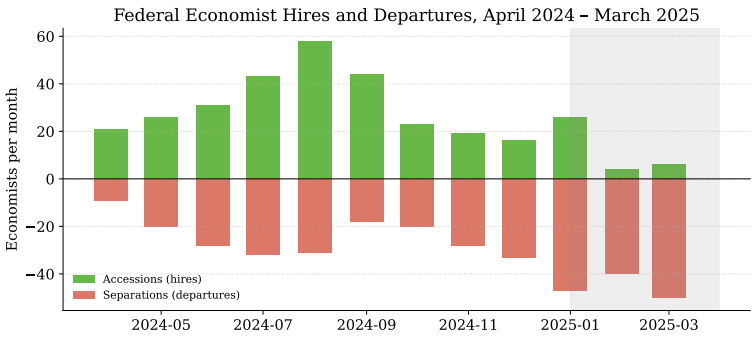
Number and Pay of Federal Economists



Source: OPM FedScope employment cubes (Sep) and the preliminary March 2025 snapshot — one consistent universe. March 2025 still counts administrative-leave / deferred-resignation staff as employed. Real pay deflated by CPI-U to March 2025 dollars.

- Headcount −2.2% (4,998 → 4,888), Sep 2024 → Mar 2025.
- Mean pay +2.8% nominal, +1.4% **real** (CPI-U) — almost entirely the January 2025 pay raise (2.0%), not a shift to more senior staff.

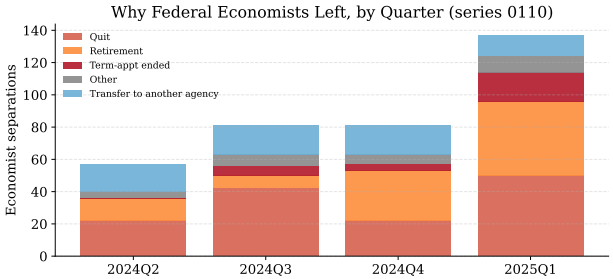
Hires and Departures: the Clearer Signal



Shaded: Jan-Mar 2025. Joiners: mean 2 yr service, $114,890. Leavers: mean 13 yr, $144,938 (mid-career departures, not juniors).

- Quarterly net: $+21$, $+64$, -23, -101 (2025 Q1). Hiring collapsed ($145 \rightarrow 36$); departures spiked ($81 \rightarrow 137$).
- Leavers averaged **12.8 years** of service and $145k — mid-career departures, not junior attrition.

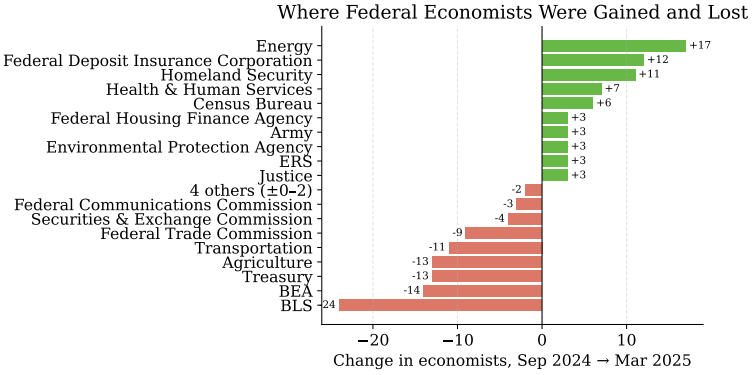
Why They Left: A Drawdown, Not Ordinary Turnover



2025 Q1 holds 86% of the year's early-out retirements, 64% of term-appointment endings, 42% of voluntary retirements and 37% of quits; transfers to other agencies fell — economists left government rather than moving within it.

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- Transfers to other agencies *fell* — economists left government rather than moving within it.

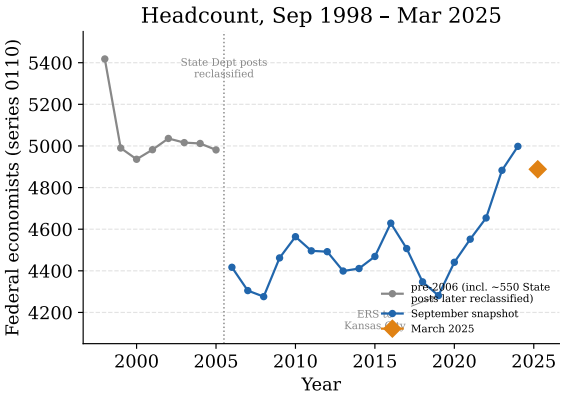
Where Economists Were Gained and Lost

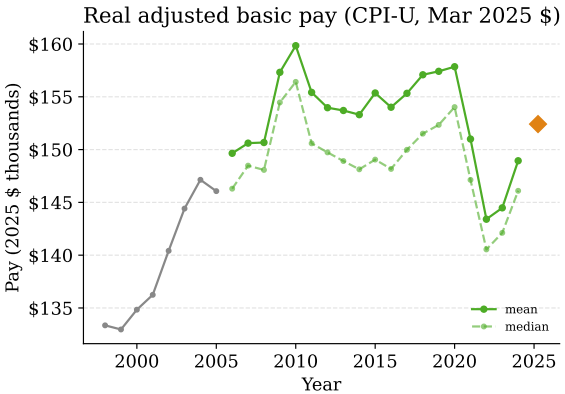


Agency × pay-plan × grade cells with ≤10 staff are suppressed (~76% coverage). Agencies changing by 0–2 are pooled.

- Statistical agencies among the losers: **BLS** −24**, BEA** −14; also Treasury −13, Agriculture −13, FTC −9.
- Gained: Energy +17, FDIC +12, DHS +11, HHS +7, Census +6.

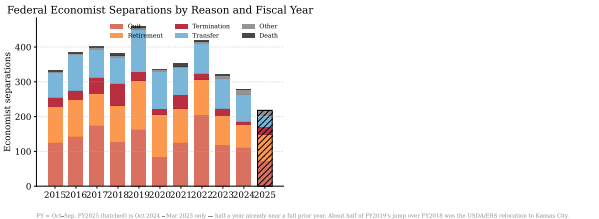
Historical Context: Economists Since 1998

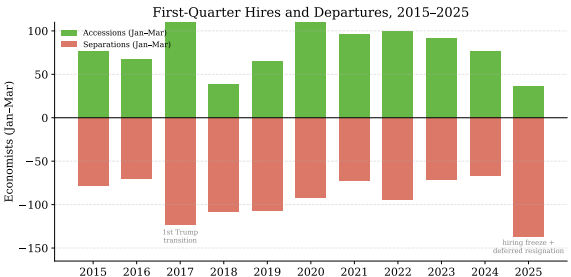




- On a consistent post-2006 basis, headcount ran 4,280–5,000. March 2025 (4,888) is still near the *top* of that range — the drop so far unwinds about a year of growth, not decades.
- Real pay is mid-range: above the late 1990s, below the 2010 and 2018–20 peaks.

Historical Context: 2025 Outflow vs. Prior Years





- FY2024 was the calmest year in the series (279 separations); FY2025's **first six months** already hit 218. Jan–Mar 2025 had the most departures (137) and the fewest hires (36) of any Q1 since 2015.
- The one precedent — FY2019 — was about half one agency (the ERS move to Kansas City). Formal RIFs are near-zero for economists in every year, so 2025's exits run through quits, retirements and term endings. The return-to-office order and the buyout offer are co-timed.

2025 Update: Robustness

Definition	Sep 2024	Mar 2025	Headcount	Pay
0110 economist (headline)	4,998	4,888	−2.2%	+2.8%
0110 + 0119 economics assistant	5,333	5,219	−2.1%	+2.8%
0110 with a doctorate (Foster proxy)	—	1,386	—	mean $179k

- Result is not sensitive to the series definition.
- Foster et al. (2020, 2023) restrict to **PhD** federal economists — ≈ the 1,386 doctorate holders here (28% of 0110). Explains their Treasury/Agriculture/HHS focus vs. our BLS-heavy full population.
- Shared universe exclusions: CEA, Federal Reserve Board, CBO, USPS, intelligence agencies.

PhD Economist Placement Data: Overview

Sample characteristics

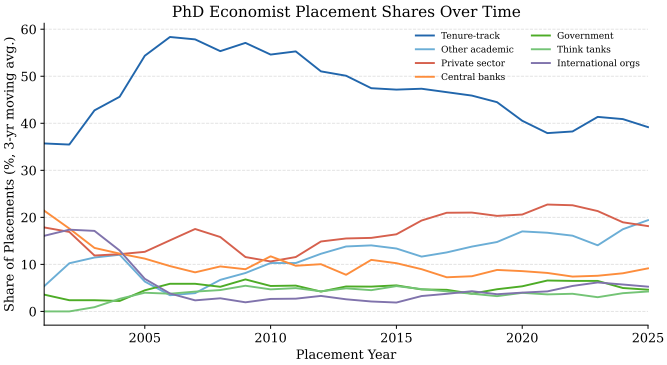
- 8,824 placements, 44 programs
- Placement years: 1991–2025
- Includes international programs: Stockholm, UBC, LSE, Sciences Po, Mannheim, EUI, CEMFI, ...
- 7 placement categories: tenure-track, other academic, private sector, central banks, government, think tanks, international orgs

Overall placement distribution

Category	Share
Tenure-track	49.4%
Private sector	18.8%
Other academic	12.2%
Central banks	7.2%
International orgs	4.6%
Think tanks	3.8%
Government	4.1%

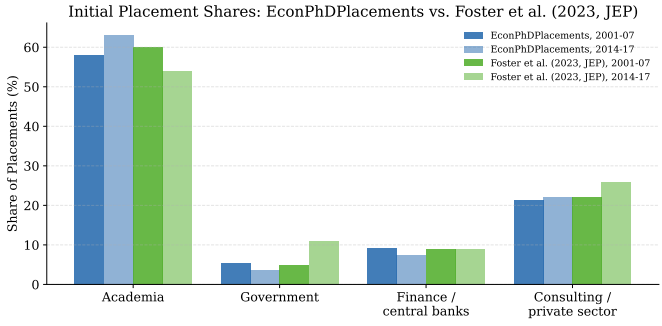
Academia total: 61.6%

Placement Trends Over Time



- Tenure-track share declining since ≈2010; other academic rising
- Private sector growing steadily; central bank placements relatively stable

Placement Shares: EconPhDPlacements vs. Foster et al. (2023, JEP)



Note: 'International orgs' (4-6% in EconPhDPlacements) not captured in paper (LEHD = US only).
Paper: Foster, McEntarfer & Sandler (2023, JEP). Govt in paper = NAICS 92 only.

Reconciling the Differences

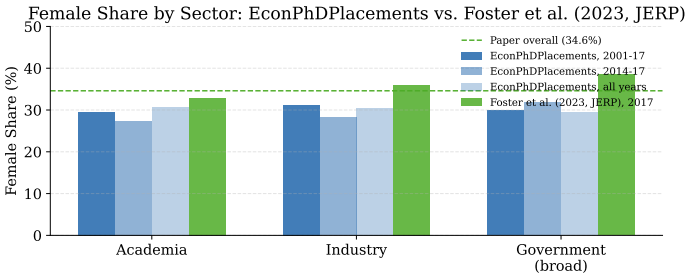
Our academia share (63%, 2014–17) is higher than paper's (54%, all programs):

- Our 44 programs skew toward **top research universities** — even for top-5 programs, paper shows 58.6%
- Placement website data = **announced** initial placements; LEHD = actual employment (captures people who planned academia but moved)
- International programs in our sample (Stockholm, UBC, ...) may have higher academic placement rates

Our government share (3.6%, 2014–17) is lower than paper's (11%):

- Top programs go to government less (paper's top-5: 8.6%; others: 11.4%)
- Government placements may be under-reported on department websites
- Paper's government includes all levels (federal + state/local, NAICS 92)

Female Share by Sector: EconPhDPlacements vs. Foster et al.



Note: Gender assigned via gender_guesser (50.5% of placements). Unassigned names skew international (likely male-heavy), so our female shares are likely understated. Paper uses self-reported SED gender (100% coverage). D&E = Diversity & Equity in Labor Market Outcomes for Economists (Foster, McEntarfer & Sandler, 2023, JERP).

Female Share in Placements: Caveats and Takeaways

Our estimates (29–32%) are below the paper's (33–38%):

- **Gender assignment gap:** only 51% assigned; unassigned names are disproportionately international and likely male-heavy $\Rightarrow$ our female shares are understated by an unknown amount
- **Initial placements vs. cross-section:** paper shows the 2017 stock of employed PhD economists (includes earlier cohorts with higher female retention); ours shows the flow of new placements
- **Ranking within sectors is preserved:** government $>$ industry $>$ academia in both datasets

Next step: use gender-specific name databases (e.g., Social Security Administration data, genderize.io) to improve coverage for international names

Summary I

Federal salary (FedsDataCenter, FY2015–2024):

- Raw median gender gap: 0–3.4%, broadly consistent with Foster et al. (2020)
- Gap is entirely between-grade, not within-grade; BEA, Census, BLS show female salary *advantage* at median
- Female share rising: 32% $\rightarrow$ 34%; gap re-widened post-2021

The 2025 update (OPM FedScope, Sep 2024 $\rightarrow$ Mar 2025):

- Headcount -2.2%; mean pay $+2.8\%$ nominal / $+1.4\%$ real (almost entirely the January 2025 raise)
- Flows show the turn: 2025 Q1 net -101, hiring collapsing as mid-career economists (mean 12.8 yr service) depart
- BLS and BEA among the agencies losing economists

Summary II

- No gender in FedScope's raw files (any year); FedsDataCenter, with name-based gender, ends at FY2024

PhD placements (econphdplacements.com, 1991–2025):

- Sample skews toward top programs; academia share higher than national average
- Government placements under-represented vs. LEHD-based paper
- International orgs (4–6%) visible here but absent from LEHD data
- Female share understated due to gender assignment coverage; ordering across sectors preserved

Next steps:

- Improve gender assignment for international names
- Link placement and salary data on individual economists
- Extend to regression-based gap estimates controlling for grade/agency

References I

- Foster, L., Manzella, J., McEntarfer, E., & Sandler, D.H. (2020). "Employment and Earnings for Federal Government Economists: Empirical Evidence by Gender and Race." *AEA Papers and Proceedings*, 110, 210–214.
- Foster, L., McEntarfer, E., & Sandler, D.H. (2023). "Diversity and Equity in Labor Market Outcomes for Economists." *Journal of Economics, Race, and Policy*.
- Foster, L., McEntarfer, E., & Sandler, D.H. (2023). "Early Career Paths of Economists Inside and Outside of Academia." *Journal of Economic Perspectives*, 37(4), 231–250.
- Data: FedsDataCenter.com (OPM federal salary data, FY2015–2024)
- Data: econphdplacements.com (PhD economist placement data, 1991–2025)